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Motor

Abstract

A motor includes a casing, a stator, and a rotor. The casing includes an inner periphery having a plurality of first assembling structures. The stator is disposed on the casing and includes a plurality of magnetic pole modules each of which is independently separable. Each of the plurality of magnetic pole modules includes at least one second assembling structure. Each of the plurality of magnetic pole modules is detachably coupled in an axial direction to the casing by coupling each of the at least one second assembling structure to an associated one of the plurality of first assembling structures. The rotor is rotatably disposed within the stator. An air gap is formed between the rotor and the stator.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4837471	12/1988	Kostoss	310/227	H02K 1/185
6927524	12/2004	Pyntikov et al.	N/A	N/A
8860287	12/2013	Longtin et al.	N/A	N/A
11394255	12/2021	Pilgrim et al.	N/A	N/A
11496008	12/2021	Georgiou et al.	N/A	N/A
11831201	12/2022	Yang et al.	N/A	N/A
11916435	12/2023	Pilgrim et al.	N/A	N/A
2007/0273239	12/2006	Kobayashi	310/216.044	H02K 1/185
2012/0133145	12/2011	Longtin	290/55	H02K 1/185
2020/0381962	12/2019	Georgiou	N/A	H02K 1/16
2022/0077727	12/2021	Pilgrim	N/A	H02K 9/19

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202663218	12/2012	CN	N/A
207926297	12/2017	CN	N/A
213585297	12/2020	CN	N/A
216489942	12/2021	CN	N/A
0313392	12/1992	EP	N/A
2003023738	12/2002	JP	N/A
2005510995	12/2004	JP	N/A
2005278333	12/2004	JP	N/A
2021164217	12/2020	JP	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The application claims the benefit of Taiwan application Ser. No. 111124978, filed on Jul. 4, 2022, and the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a motor and, more particularly, to a motor permitting independent replacement of a magnetic pole module.

2. Description of the Related Art

(3) A conventional motor includes a stator and a rotor. The stator is assembled on a casing of the motor. The stator includes a plurality of iron cores and a coil unit. Each iron core is integrally formed by pressing to form an annular member. The coil unit is wound around the plurality of iron cores forming the annular member. Each iron core includes a magnetic yoke and a pole shoe. Each two adjacent magnetic yokes are connected with each other. Each two adjacent pole shoes have a spacing therebetween. The rotor is rotatably mounted in a space delimited by the pole shoes. An air gap is formed between an outer periphery of the rotor and the pole shoes of the stator, so that the rotor can rotate smoothly relative to the stator.

(4) In the above conventional motor, since each iron core is integrally formed by pressing to form an annular member, after the coil unit is wound around each iron core, the iron cores wound by the coil units are assembled to the casing. However, if the size of the above conventional motor is larger, the size of the mold for the integrally formed iron cores will also be large, resulting in higher costs for the iron core mold. Furthermore, manufacturing is not easy, as wire winding of the integrally formed iron cores is required. Moreover, after the above conventional motor has been used for a period of time, when the stator and the casing become locally loose after long-term operation of the motor, the iron cores which form an annular member as a whole must be adjusted and assembled again. Alternatively, in a case that any one of the coil units of the stator is damaged, the iron cores which form an annular member as a whole must be detached in order to remove the damaged coil unit from the iron cores, resulting in inconvenience in assembly and disassembly.

(5) Thus, it is necessary to improve the conventional motor.

SUMMARY OF THE INVENTION

(6) To solve the above problem, it is an objective of the present invention to provide a motor which permits removal of a single magnetic pole module for replacement or maintenance while permitting easy assembly and disassembly.

(7) It is another objective of the present invention to provide a motor which can reduce the manufacturing costs.

(8) It is a further objective of the present invention to provide a motor which permits stable coupling between the stator and the casing.

(9) It is still another objective of the present invention to provide a motor which permits easy removal of the magnetic pole modules

(10) As used herein, the term “a”, “an” or “one” for describing the number of the elements and members of the present invention is used for convenience, provides the general meaning of the scope of the present invention, and should be interpreted to include one or at least one.

Furthermore, unless explicitly indicated otherwise, the concept of a single component also includes the case of plural components.

(11) As used herein, the term “engagement”, “coupling”, “assembly”, or similar terms is used to include separation of connected members without destroying the members after connection or inseparable connection of the members after connection. A person having ordinary skill in the art

would be able to select according to desired demands in the material or assembly of the members to be connected.

(12) A motor according to the present invention includes a casing, a stator, and a rotor. The casing includes an inner periphery having a plurality of first assembling structures. The stator is disposed on the casing and includes a plurality of magnetic pole modules each of which is independently separable. Each of the plurality of magnetic pole modules includes at least one second assembling structure. Each second assembling structure is detachably coupled in an axial direction to an associated one of the plurality of first assembling structures. The rotor is rotatably disposed within the stator. An air gap is formed between the rotor and the stator.

(13) Thus, in the motor according to the present invention, the stator includes a plurality of independently separable magnetic pole modules each having at least one second assembling structure. Each magnetic pole module may be axially and detachably coupled to the casing by coupling each second assembling structure to an associated first assembling structure of the casing. When one of the plurality of magnetic pole modules of the stator is damaged, the damaged magnetic pole module may be removed alone for replacement or maintenance. After maintenance, by the provision of each second assembling structure and the associated first assembling structure, the repaired magnetic pole module can be easily reassembled to the casing, which may avoid adverse influence on the arrangement of the other magnetic pole modules near the repaired magnetic pole module, permitting easy assembly and disassembly.

(14) In an example, each of the plurality of magnetic pole modules may include an outer face and an inner face. Each outer face may be adjacent to an associated one of the plurality of first assembling structures. Each inner face may be remote from the associated first assembling structure. Thus, the second assembling structures of the plurality of magnetic pole modules can be easily coupled with the plurality of first assembling structures, which is convenient in assembly.

(15) In an example, each of the plurality of first assembling structures may include a post having two axial end faces opposite to each other. The post of each of the plurality of first assembling structures may include two guiding grooves extending through the two axial end faces. Each second assembling structure may include a protrusion which can be engaged with an associated one of the two guiding grooves of an associated first assembling structure. Thus, each of the plurality of magnetic pole modules can be assembled between two adjacent first assembling structures, which is convenient in assembly.

(16) In an example, the post of each of the plurality of first assembling structures may include two engaging holes respectively formed on the two axial end faces. The two engaging holes may be located between the two guiding grooves. Thus, the structure is simple and easy to manufacture, thereby reducing the manufacturing costs.

(17) In an example, the casing may include a first casing part formed of a plurality of arcuate blocks connected to each other. The plurality of arcuate blocks may together tightly clamp the plurality of magnetic pole modules of the stator. Thus, the plurality of magnetic pole modules can avoid from becoming loose, such that the plurality of magnetic pole modules of the stator can be stably coupled.

(18) In an example, each of the plurality of arcuate blocks may include two lugs. Each of the two lugs of each of the plurality of arcuate blocks is aligned with one of the two lugs of an adjacent arcuate block. Each pair of aligned lugs is coupled by a fastening unit. Thus, the structure is simple and easy to assemble.

(19) In an example, each of the plurality of first assembling structures may include two channels and a limiting column formed between the two channels. Each second assembling structure may include two hooked columns and a limiting groove formed between the two hooked columns. The two hooked columns may engage with the two channels. The limiting column may engage with the limiting groove. Thus, by the provision of the second assembling structures, each of the plurality of magnetic pole modules can be detachably coupled between two adjacent first assembling structures

in the axial direction, which is convenient in assembly.

(20) In an example, each of the plurality of first assembling structures may include a slot and a pin. The pins may be disposed on a first casing part and a second casing part of the casing. Each second assembling structure may include a hollow column. Each hollow column may be aligned with an associated pin. Thus, during assembly, the hollow column of each second assembling structure axially engages with an associated pin of the first casing part aligned with the hollow column. Then, the pins of the second casing part axially engage with the hollow columns of the second assembling structures aligned with the pins, such that the pins respectively extend into the hollow columns of the second assembling structures, thereby stably coupling the casing with the stator.

(21) In an example, each pin located on the first casing part may be received in an associated slot. Thus, the structure is simple and easy to manufacture, thereby reducing the manufacturing costs.

(22) In an example, each pin located on the first casing part may include an end connected to the first casing part and another end facing the second casing part. Each pin located on the second casing part may include an end connected to the second casing part and another end facing the first casing part. Thus, the structure is simple and easy to manufacture, thereby reducing the manufacturing costs.

(23) In an example, the motor may further include a plurality of fixing units axially coupled with the casing and the stator. Thus, the plurality of magnetic pole modules can be stably coupled to the casing, such that the stator and the casing can be stably coupled.

(24) In an example, each of the plurality of fixing units may include a pressing member and a positioning member. Each pressing member may include a padding portion and two pressing portions. The two pressing portions are connected to the padding portion and press against an upper face or a lower face of an associated second assembling structure. The positioning member may extend through the padding portion and engage with an engaging hole of an associated one of the plurality of first assembling structures. Thus, it is assured that the plurality of magnetic pole modules can be stably coupled with the casing to effectively prevent the plurality of magnetic pole modules from displacing relative to or falling from the casing.

(25) In an example, each of the plurality of fixing units may include a pressing member and a positioning member. Each pressing member may include a padding board portion and a pressing board portion. A connecting board may be connected between the padding board portion and the pressing board portion. Each pressing board portion presses against an upper face or a lower face of an associated second assembling structure. The positioning member may extend through the padding board portion and engage with a mounting hole of the casing. Thus, it is assured that the plurality of magnetic pole modules can be stably coupled with the casing to effectively prevent the plurality of magnetic pole modules from displacing relative to or falling from the casing.

(26) In an example, the motor may further include a plurality of fasteners. The plurality of fasteners may extend through the casing and press against the plurality of magnetic pole modules, respectively. Thus, after assembly, the stator will not rotate relative to the casing to effectively prevent the plurality of magnetic pole modules from displacing relative to or falling from the casing.

(27) In an example, the casing may include a plurality of radial through-holes. The plurality of fasteners may extend through and engage with the plurality of radial through-holes, respectively. Thus, the structure is simple and easy to manufacture, thereby reducing the manufacturing costs.

(28) In an example, the motor may further include a plurality of blocks. Each of the plurality of blocks is disposed on an associated one of the plurality of magnetic pole modules. A push member extends through a through-hole of the casing and is configured to push one of the plurality of blocks. Thus, the plurality of magnetic pole modules can be removed from the casing in the axial direction, thereby permitting easy removal of the plurality of magnetic pole modules.

(29) In an example, each of the plurality of magnetic pole modules may include a plurality of receiving pits respectively located on an upper face and/or a lower face of the associated magnetic

pole module. Each of the plurality of blocks may be located in one of the plurality of receiving pits. Thus, the push member can push the block from above or below, thereby permitting easy disassembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will become more fully understood from the detailed description given hereinafter and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present invention, and wherein:
- (2) FIG. 1 is an exploded, perspective view of a motor of a first embodiment according to the present invention.
- (3) FIG. 2 is a partial, exploded, perspective view of the motor of the first embodiment according to the present invention.
- (4) FIG. 3 is a cross sectional view of the motor of the first embodiment according to the present invention after assembly.
- (5) FIG. 4 is a cross sectional view taken along section line 4-4 of FIG. 3.
- (6) FIG. 5 is a cross sectional view taken along section line 5-5 of FIG. 4.
- (7) FIG. 6 is a cross sectional view taken along section line 6-6 of FIG. 4.
- (8) FIG. 7 is a cross sectional view similar to FIG. 6, illustrating disassembly of a magnetic pole module of the motor of the first embodiment according to the present invention.
- (9) FIG. 8 is an exploded, perspective view of a motor of a second embodiment according to the present invention.
- (10) FIG. 9 is an exploded, perspective view of a motor of a third embodiment according to the present invention.
- (11) FIG. 10 is a top view of the motor of the third embodiment according to the present invention.
- (12) FIG. 11 is a cross sectional view taken along section line 11-11 of FIG. 10.
- (13) FIG. 12 is an exploded, perspective view of a motor of a fourth embodiment according to the present invention.
- (14) When the terms “front”, “rear”, “left”, “right”, “up”, “down”, “top”, “bottom”, “inner”, “outer”, “side”, and similar terms are used herein, it should be understood that these terms have reference only to the structure shown in the drawings as it would appear to a person viewing the drawings and are utilized only to facilitate describing the invention, rather than restricting the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (15) With reference to FIG. 1, a motor of a first embodiment according to the present invention includes a casing 1, a stator 2, and a rotor 3. The stator 2 is disposed on the casing 1. The rotor 3 is rotatably disposed within the stator 2.
- (16) With reference to FIGS. 1 and 2, the casing 1 may include a first casing part 1a and a second casing part 1b. The second casing part 1b may be coupled with the first casing part 1a to jointly envelope the stator 2. The coupling between the first casing part 1a and the second casing part 1b may be buckling, threading connection, welding, etc. The present invention is not limited in this regard. The casing 1 may include an axial hole H which may be located on the second casing part 1b. An inner periphery N of the casing 1 includes a plurality of first assembling structures 11 which may be located on the first casing part 1a. The plurality of first assembling structures 11 is spaced from each other and extends in an axial direction of the axial hole H. Each of the plurality of first assembling structures 11 may include a post 11a which may protrude from the inner periphery N of the casing 1. Each of the plurality of first assembling structures 11 may include two axial end faces 11b opposite to each other. The two axial end faces 11b are respectively located on top and bottom

faces of the post **11a**. The post **11a** of each of the plurality of first assembling structures **11** includes two guiding grooves **11c** extending through the two axial end faces **11b**. The post **11a** of each of the plurality of first assembling structures **11** may further include two engaging holes **11d**. The two engaging holes **11d** may be respectively formed on the two axial end faces **11b** and may or may not intercommunicate with each other. The present invention is not limited in this regard. In this embodiment, the two engaging holes **11d** may extend through the two axial end faces **11b**, such that the two engaging holes **11d** intercommunicate with each other. Furthermore, the two engaging holes **11d** may be located between the two guiding grooves **11c**.

(17) With reference to FIGS. **1**, **2**, and **6**, the case **1** may include a plurality of radial through-holes **12** located on the first casing part **1a**. Each of the plurality of radial through-holes **12** may be located between two adjacent first assembling structures **11**. The casing **1** may include a plurality of through-holes **13**. The plurality of through-holes **13** may be located on the first casing part **1a** and the second casing part **1b**, respectively. In this embodiment, each of the plurality of through-holes **13** is in the form of a screw hole for purpose of explanation.

(18) With reference to FIG. **2**, the stator **2** may include a plurality of magnetic pole modules **2a** each of which can be independently separated. The motor may include plural-phase coils. As an example, the motor of this embodiment is a three-phase motor. Each of the plurality of magnetic pole modules **2a** may include three-phase winding coils (not shown). Each of the plurality of magnetic pole modules **2a** includes an outer face **F1** and an inner face **F2**. Each outer face **F1** is the face adjacent to an associated one of the plurality of first assembling structures **11**. Each inner face **F2** is the face remote from the associated first assembling structure **11**. Each of the plurality of magnetic pole modules **2a** includes an upper face **F3** and a lower face **F4** opposite to the upper face **F3**. Each of the plurality of magnetic pole modules **2a** includes at least one second assembling structure **21**. In this embodiment, each of the plurality of magnetic pole modules **2a** includes two second assembling structures **21** respectively located on two sides of the magnetic pole module **2a**. The two second assembling structures **21** are located between the outer face **F1** and the inner face **F2** and are nearer to the outer face **F1**. The two second assembling structures **21** are respectively aligned with two adjacent first assembling structures **11**. Each second assembling structure **21** is detachably coupled to the two adjacent first assembling structures **11** in the axial direction. In this embodiment, each second assembling structure **21** may include a protrusion **21a** which may be engaged with an associated one of the two guiding grooves **11c** of an associated first assembling structure **11**, such that the associated magnetic pole module **2a** may be assembled between two adjacent first assembling structures **11**. Furthermore, each of the plurality of magnetic pole modules **2a** may include a plurality of receiving pits **22** respectively located on the upper face **F3** and/or the lower face **F4**. The plurality of receiving pits **22** on the same magnetic pole module **2a** may be located between the two second assembling structures **21**, and the plurality of receiving pits **22** may be aligned with the plurality of through-holes **13** of the casing **1**, respectively.

(19) With reference to FIGS. **1**, **3**, and **4**, the rotor **3** may include a shaft **31** and a magnet portion **32**. An end of the shaft **31** passes through the axial hole **H** of the casing **1** and extends beyond the second casing part **1b**. The magnet portion **32** is disposed around the shaft **31**. The magnet portion **32** may be, e.g., coupled to the shaft **31** via a base, and the base includes a plurality of annularly disposed magnets, or any other equivalent structures. The present invention is not limited in this regard. The magnet portion **32** of the rotor **3** and the inner face **F2** of the plurality of magnetic pole modules **2a** of the stator **2** have an air gap **G** formed therebetween. Thus, the magnet portion **32** may proceed with magnetic induction with the plurality of magnetic pole modules **2a** to drive the shaft **31** to rotate relative to the stator **2**.

(20) With reference to FIGS. **2** and **5**, the motor according to the present invention further includes a plurality of fixing units **4** axially coupled with the casing **1** and the stator **2**. Each of the plurality of fixing units **4** may include a pressing member **41** and a positioning member **42**. Each pressing member **41** may include a padding portion **411** and two pressing portions **412**. Each padding

portion **411** may be coupled with one of the two axial end faces **11b** of an associated first assembling structure **11**. In this embodiment, the padding portion **411** does not abut the axial end face **11b** of the first assembling structure **11**. In another embodiment, the padding portion **411** may abut an associated axial end face **11b** of the first assembling structure **11**. The two pressing portions **412** are connected to the padding portion **411**, extend into the two guiding grooves **11c** of the first assembling structure **11**, and press against an upper face **F3** or a lower face **F4** of an associated second assembling structure **21**. The positioning member **42** extends through the padding portion **411** and engages with an engaging hole **11d** of an associated first assembling assembly **11**, such that the plurality of magnetic pole modules **2a** can be stably coupled with the casing **1**, thereby effectively preventing the plurality of magnetic pole modules **2a** from displacing relative or falling from the casing **1**.

(21) With reference to FIGS. **2** and **3**, the motor according to the present invention may further comprise a plurality of fasteners **5**. The plurality of fasteners **5** respectively extends through the plurality of radial through-holes **12** of the casing **1**. The plurality of fasteners **5** may be in threading connection with the plurality of radial through-holes **12**. Furthermore, the plurality of fasteners **5** may press against the outer faces **F1** of the plurality of magnetic pole modules **2a**, respectively, such that the plurality of magnetic pole modules **2a** may be stably coupled with the casing **1**. As a result, the stator **2** after assembly will not rotate relative to the casing **1**, thereby effectively preventing the plurality of magnetic pole modules **2a** from displacing relative to or falling from the casing **1**.

(22) With reference to FIGS. **6** and **7**, the motor according to the present invention may further include a plurality of blocks **6**. Each of the plurality of blocks **6** may be made of metal. Each of the plurality of blocks **6** may be located in one of the plurality of receiving pits **22** of an associated magnetic pole module **2a**, such that the plurality of blocks **6** may be aligned with the plurality of through-holes **13**, respectively. Each of the plurality of blocks **6** may be engaged in an associated receiving pit **22** by integral molding injection to permit easy manufacturing. Therefore, a push member **T** may be used to extend through one of the plurality of through-holes **13** of the casing **1** and to push one of the plurality of blocks **6**, such that the associated magnetic pole module **2a** may displace axially relative to the casing **1** to permit removal of the magnetic pole module **2a**, which permits easy maintenance or replacement of the magnetic pole module **2a**. It is particularly noted that each push member **T** may be in the form of a screw to be in threading connection with the associated through-hole **13**, permitting the push member **T** to easily push the associated push block **6**.

(23) With reference to FIGS. **2** and **3**, when the stator **2** is energized, the plurality of magnetic pole modules **2a** of the stator **2** and the magnetic portion **32** form repulsive magnetic poles to drive the rotor **3** to rotate. After the motor according to the present invention has been used for a period of time, since the stator **2** includes a plurality of independently separable magnetic pole modules **2a** and since each second assembling structure **21** is detachably coupled to an associated one of the plurality of first assembling structures **11** in the axial direction, when one of the plurality of magnetic pole modules **2a** of the stator **2** is damaged, the damaged magnetic pole module **2a** can be removed alone, and replacement or maintenance can be achieved by simply removing the coil unit (not shown) on the damaged magnetic pole unit **2a**. After maintenance, the repaired magnetic pole module **2a** is remounted to the casing **1**. By the provision of each second assembling structure **21** detachably coupled to an associated first assembling structure **11** in the axial direction, adverse influence on the size and location of the magnetic pole modules **2a** near the repaired magnetic pole module **2a** can be avoided, permitting easy assembly and disassembly.

(24) Please refer to FIG. **8** showing a motor of a second embodiment according to the present invention. The second embodiment of the present invention is substantially the same as the first embodiment. In the second embodiment, the casing **1** includes a first casing part **1a** formed of a plurality of arcuate blocks **14** connected to each other. Each of the plurality of arcuate blocks **14**

includes two lugs **141**. Each of the two lugs **141** of each of the plurality of arcuate blocks **14** is aligned with one of the two lugs **141** of an adjacent arcuate block **14**. Each pair of aligned lugs **141** is coupled by a fastening unit **15**. Thus, the plurality of arcuate blocks **14** may together tightly clamp the plurality of magnetic pole modules **2a** of the stator **2**, further avoiding the plurality of magnetic pole modules **2a** from becoming loose.

(25) Please refer to FIGS. **9-11** showing a motor of a third embodiment according to the present invention. The third embodiment of the present invention is substantially the same as the first embodiment. In the third embodiment, each of the plurality of first assembling structures **11** may include two channels **11e** which may be recessed in the inner periphery N of the casing **1**. A limiting column **11f** may be formed between the two channels **11e**. Each limiting column **11f** may be substantially T-shaped. Furthermore, each second assembling structure **21** of the stator **2** may be formed on the outer face F**1** of the associated magnetic pole module **2a**. Each second assembling structure **21** may include two hooked columns **21b** which may protrude beyond the outer face F**1**. A limiting groove **21c** may be formed between the two hooked columns **21b**. The two hooked columns **21b** may engage with the two channels **11e** of an associated first assembling structure **11**. The limiting column **11f** may engage with the limiting groove **21c** of an associated second assembling structure **21**. Therefore, each of the plurality of magnetic pole modules **2a** may be axially and detachably coupled to two adjacent first assembling structures **11** by the two second assembling structures **21**. As a result, by providing each second assembling structure **21** axially and detachably coupled to an associated first assembling structure **11**, when one of the plurality of magnetic pole modules **2a** of the stator **2** is damaged, the damaged magnetic pole module **2a** may be removed alone for replacement or maintenance.

(26) Furthermore, each pressing member **41** may include a padding board portion **413** and a pressing board portion **414**. A connecting board **415** is connected between the padding board portion **413** and the pressing board portion **414**. In this embodiment, the pressing member **41** may be in the form of a Z-shaped piece. Each padding board portion **413** may engage one of the two axial end faces **11b** of an associated first assembling structure **11**. In this embodiment, each padding board portion **413** abuts an axial end face **11b** of an associated first assembling structure **11** for purpose of explanation. In another embodiment, each padding board portion **413** may not abut the axial end face **11b** of the associated first assembling structure **11**. Each pressing board portion **414** may press against an upper face F**3** or a lower face F**4** of an associated second assembling structure **21**. The positioning member **42** may extend through the padding board portion **413** and engage with a mounting hole **16** of the casing **1**, such that the plurality of magnetic pole modules **2a** can be stably coupled to the casing **1**, thereby effectively preventing the plurality of magnetic pole modules **2a** from displacing relative to or falling from the casing **1**.

(27) Please refer to FIG. **12** showing a motor of a fourth embodiment according to the present invention. The fourth embodiment of the present invention is substantially the same as the third embodiment. In the fourth embodiment, each of the plurality of first assembling structures **11** may include a slot **11g** and a pin **11h**. Each slot **11g** may be recessed in the inner periphery N of the casing **1**. The pins **11h** may be disposed on the first casing part **1a** and the second casing part **1b**. Each pin **11h** located on the first casing part **1a** includes an end connected to the first casing part **1a** and another end facing the second casing part **1b**. Each pin **11h** on the first casing part **1a** is located in an associated slot **11g**. Furthermore, each pin **11h** located on the second casing part **1b** includes an end connected to the second casing part **1b** and another end facing the first casing part **1a**. Moreover, the pins **11h** on the first casing part **1a** are misaligned from the pins **11h** on the second casing part **1b**. Each second assembling structure **21** of the stator **2** is located on the outer face F**1** of an associated magnetic pole module **2a**. Each second assembling structure **21** may include a hollow column **21d**. Each hollow column **21d** may be aligned with an associated pin **11h**. In assembly, the hollow column **21d** of the second assembling structure **21** is axially coupled with a pin **11h** of the first casing part **1a** aligned with the hollow column **21d**. Then, the pin **11h** of the

second casing part **1b** is axially coupled with a hollow column **21d** of the second assembling structure **21** aligned with the pin **11h**, such that the pins **11h** may respectively extend into the hollow columns **21d** of the second assembling structures **21**. Therefore, by the provision of each second assembling structure **21** axially and detachably coupled to an associated first assembling structure **11**, when one of the plurality of magnetic pole modules **2a** of the stator **2** is damaged, the damaged magnetic pole module **2a** may be removed alone for replacement or maintenance.

(28) In summary, in the motor according to the present invention, the stator includes a plurality of independently separable magnetic pole modules each having at least one second assembling structure. Each magnetic pole module is axially and detachably coupled to the casing by coupling each second assembling structure to an associated first assembling structure of the casing. When one of the plurality of magnetic pole modules of the stator is damaged, the damaged magnetic pole module may be removed alone for replacement or maintenance. After maintenance, by the provision of each second assembling structure and the associated first assembling structure, the repaired magnetic pole module can be easily reassembled to the casing, which may avoid adverse influence on the arrangement of the other magnetic pole modules near the repaired magnetic pole module, permitting easy assembly and disassembly.

(29) Although the invention has been described in detail with reference to its presently preferable embodiments, it will be understood by one of ordinary skill in the art that various modifications can be made without departing from the spirit and the scope of the invention, as set forth in the appended claims.

Claims

1. A motor comprising: a casing including an inner periphery having a plurality of first assembling structures; a stator disposed on the casing and including a plurality of magnetic pole modules, wherein each of the plurality of magnetic pole modules is independently separable, wherein each of the plurality of magnetic pole modules includes at least one second assembling structure, and wherein each of the plurality of magnetic pole modules is detachably coupled in an axial direction to the casing by coupling each of the at least one second assembling structure to an associated one of the plurality of first assembling structures; a rotor rotatably disposed within the stator, wherein an air gap is formed between the rotor and the stator; and a plurality of fixing units axially coupled with the casing and the stator; wherein each of the plurality of first assembling structures includes a post having two axial end faces opposite to each other, wherein the post of each of the plurality of first assembling structures includes two guiding grooves extending through the two axial end faces, wherein each second assembling structure includes a protrusion engaged with an associated one of the two guiding grooves of an associated first assembling structure; wherein each of the plurality of fixing units includes a pressing member and a positioning member, wherein each pressing member includes a padding portion and two pressing portions, wherein the two pressing portions are connected to the padding portion, extend into the two guiding grooves of the first assembling structure, and press against an upper face or a lower face of an associated second assembling structure, and wherein the positioning member extends through the padding portion and engages with an engaging hole of an associated one of the plurality of first assembling structures.

2. The motor as claimed in claim 1, wherein each of the plurality of magnetic pole modules includes an outer face and an inner face, wherein each outer face is adjacent to an associated one of the plurality of first assembling structures, and wherein each inner face is remote from the associated first assembling structure.

3. The motor as claimed in claim 1, wherein the post of each of the plurality of first assembling structures includes two engaging holes respectively formed on the two axial end faces, and wherein the two engaging holes are located between the two guiding grooves.

4. The motor as claimed in claim 1, wherein the casing includes a first casing part formed of a

plurality of arcuate blocks connected to each other, and wherein the plurality of arcuate blocks together tightly clamp the plurality of magnetic pole modules of the stator.

5. The motor as claimed in claim 4, wherein each of the plurality of arcuate blocks includes two lugs, wherein each of the two lugs of each of the plurality of arcuate blocks is aligned with one of the two lugs of an adjacent arcuate block, and wherein each pair of aligned lugs is coupled by a fastening unit.

6. The motor as claimed in claim 1, further comprising a plurality of fasteners, wherein the plurality of fasteners extends through the casing and presses against the plurality of magnetic pole modules, respectively.

7. The motor as claimed in claim 6, wherein the casing includes a plurality of radial through-holes, and wherein the plurality of fasteners extends through and engages with the plurality of radial through-holes, respectively.

8. The motor as claimed in claim 1, further comprising a plurality of blocks, wherein each of the plurality of blocks is disposed on an associated one of the plurality of magnetic pole modules, and wherein a push member extends through a through-hole of the casing and is configured to push one of the plurality of blocks.

9. The motor as claimed in claim 8, wherein each of the plurality of magnetic pole modules includes a plurality of receiving pits respectively located on an upper face and/or a lower face of the associated magnetic pole module, and wherein each of the plurality of blocks is located in one of the plurality of receiving pits.
